

# Data Retention & Decommissioning Policy

Sentinel Nexus — AI Data Lifecycle

RW

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## Purpose

Per NIST AI 600-1 **GV-1.7-002** (decommissioning factors), **MS-2.10-001** (membership-inference / training-data extraction), and **MS-2.2-004** (privacy-enhancing techniques), this policy governs retention, expiry, and secure decommissioning of the AI-relevant data stores in Sentinel Nexus. It addresses control item **NC-4**.

## Data stores in scope

| Store                                    | Contents                                                    | Sensitivity |
|------------------------------------------|-------------------------------------------------------------|-------------|
| Cold storage (Parquet on MinIO/S3)       | Normalized sensor telemetry (Hive-partitioned)              | High        |
| Qdrant vector index                      | UEBA math vectors + payloads (per-investigation)            | High        |
| <code>nexus_swarm_memory</code> (Qdrant) | Verdict/immunity signatures + embeddings + incident reports | High        |
| RSI / calibration / bias-audit ledgers   | Cycle outcomes, calibration points, audit reports           | Moderate    |
| MLOps staging / training corpora         | Synthetic + sanitized telemetry SFT records                 | Moderate    |

## Retention & expiry

| Store                           | Retention                                                                 | Mechanism                                                                                                                      |
|---------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Cold storage                    | Per data-classification schedule (operator-set)                           | S3 lifecycle / partition pruning                                                                                               |
| DR snapshots                    | 7 days local                                                              | <code>dr_snapshot</code> timer                                                                                                 |
| <code>nexus_swarm_memory</code> | <b>TTL-bounded</b> (default 30 d, <code>NEXUS_MEMORY_TTL_SECONDS</code> ) | A point past TTL no longer grants auto-dismissal (control implemented; recall delegates to <code>memory_is_actionable</code> ) |
| immunity points                 |                                                                           | Append-only files                                                                                                              |
| Calibration / bias ledgers      | Rolling; archived not deleted                                             |                                                                                                                                |

**Membership-inference posture (MS-2.10-001).** `nexus_swarm_memory` stores embeddings of alert signatures. Risk: an adversary with read access could infer prior incidents. Mitigations: the store is inside the access-controlled enclave (no external exposure); the signature is keyed on stable *pattern identity* (sensor|source\_type|vector), not raw PII; the TTL bounds the window. **Open action (POA&M-4):** add a periodic

membership-inference review and evaluate differential-privacy / anonymization on the memory vectors.

## Decommissioning (GV-1.7-002)

When a model, sensor source, or the system is retired, the following are verified:

1. **Data retention requirements** — confirm legal/operational holds before purge.
2. **Secure erasure** — cryptographic erase of weights/secrets; purge of the relevant `nexus_swarm_memory` signatures and cold-storage partitions; revoke Vault paths.
3. **Data leakage after decommission** — confirm no residual in DR snapshots beyond the retention window; rotate the `integrity_secret` and any shared keys.
4. **Dependencies** — check upstream/downstream IoT/AI/data dependencies before removal (e.g. a retired sensor's `source_type` routing and SIEM fanout mapping).
5. **Open-source / model artifacts** — remove from the model registry per the N=3 + referenced-version retention rule.
6. **Operator entanglement** — communicate deactivation, reasons, and alternative process to affected operators (per MG-2.4-001).

## Privacy-enhancing handling

- Training corpora are **credential-scrubbed** at staging (`01_spool_chatml`, `01_spool_datasets`) — passwords/keys/tokens masked before any record enters the corpus.
- Outbound DLP scrubs RFC-1918 ranges and high-entropy secrets before any frontier egress; the cognitive sanitizer wraps adversary-controlled data as untrusted.
- TEVV / report retention follows the document-retention rule (GV-1.5-003): raw `.log` files are not committed; structured `.md/.xml` reports are retained for audit.

## Review

Reviewed annually and on any change to a data store's contents, classification, or the external interconnection set.